

The EIC and ePIC for a polarized-lithium programme: beam and detector parameters, their provenance, and the lithium case derived

C. Peng and J. Zhou

Physics Division, Argonne National Laboratory

Writing assisted by Claude (Anthropic)

2026-08-27 · rewritten 2026-08-28 with the far-forward optics re-derived per configuration, restated 2026-08-29 on the per-configuration transport · the specification behind Reports 1, 2 and 4 · every entry carries its source and its **status** — design value, requirement, measured, derived here, or unknown · figures from `evgen/scripts/eic_beam_figures.py`, tables from `polli_fastsim.beams`, `polli_fastsim.farforward` and `fastsim/scripts/tagging_acceptance.py`

Abstract. Every projection of the polarized-lithium programme rests on parameters that belong to the accelerator or to the detector. This report collects them, states what each one is — a design value, a requirement, a measurement, a derivation of ours, or an unknown — and derives the lithium case where nothing is published. Three results carry the rest of the programme. Ion energies are set by the revolution period, not by rigidity: ions are γ -matched to the proton configurations, so the ^6Li menu is 41 and 99–138 GeV/u with nothing between. The beam divergence, which every far-forward acceptance is an exponential in the square of, follows from the Yellow Report's proton tables with a species factor that is unity wherever the ion is γ -matched and $\sqrt{2}$ only at the rigidity-capped top: 220/380, 180/180 and 92/92 μrad (h/v) for ^6Li at the three configurations. Under those optics the intact- ^6Li recoil cannot be tagged at any configuration (9×10^{-10} to 6×10^{-27} with the measured pot aperture) and the ^6Li α spectator only through its off-rigidity window and the over-rigid inner branch beside it (1.7–2.6%); a lithium tagging optics — the horizontal β^* raised 46–164 $\times$ with the pots following the envelope, at 1/6.8–1/12.8 of the luminosity — recovers a 25–37% coherent tag and a 22–31% α tag, and is the one *new* optics this programme asks for; ^7Li keeps the published one, for which it pays nothing, so the ask is two run configurations rather than two lattice designs. The detector model is a requirement where a requirement exists and a labelled stand-in where none does: the track angular resolution and the electron-identification curve have no published ePIC source.

1 · Scope and status vocabulary

Report 1 presents the measurement and Report 2 the simulation and reconstruction chain; this report is the specification both are conditional on, in a form that separates what the machine and detector documents state from what this programme has had to derive or assume. Five status words are used throughout: **design value**, a parameter that appears in a design document; **requirement**, a specification a subsystem must meet, not a demonstrated performance; **measured**, from test beam, operations, or — for the pot aperture — a simulation of the detector geometry; **derived**, computed here from published inputs with the derivation shown; and **unknown**, not published anywhere we could reach. §2 gives the collider and the ion energy menu, §3 the beams, §4 the lithium case, §5 the far-forward acceptance of the lithium tags at each optics, §6 the ePIC detector, and §7 the list of what the programme assumes on the machine's and the detector's behalf, with owners.

2 · The collider and the ion energy menu

The EIC is two rings in the RHIC tunnel: the Electron Storage Ring at 5, 10 and 18 GeV and the Hadron Storage Ring carrying protons to 275 GeV and ions to the same magnetic rigidity, 917.3 T·m. ePIC sits at IP6; a second interaction region at IP8 is designed but not funded. The crossing angle is 25 mrad at IP6 and 35 mrad at IP8, recovered by crab cavities [1,2].

Colliding bunches requires equal revolution periods. The electrons are ultrarelativistic at every energy, so their period is fixed; the hadrons must match it, and since the circumference is adjustable only within a narrow range, the hadron velocity — hence γ — is quantised by the ring geometry, and the magnets supply whatever rigidity that γ implies for the species in the ring, up to the cap. Two limits therefore bind at the two ends of the menu: at the top the magnets bind and the ion is

rigidity-capped at $p/\text{nucleon} = 275 \text{ GeV} \times Z/A$; at the bottom the circumference binds and the ion is γ -matched to the proton configuration, at a per-nucleon energy that rigidity scaling does not give. EPIOS describes the mechanism [2]: a radial shift of up to $\pm 20 \text{ mm}$ in the arcs covers $118 < \gamma < 293$, and a bypass arc 90 cm shorter reaches $\gamma = 43.5$. The check that settles the rule is Yellow Report Table 10.2, which lists gold at 41 GeV/u — $\gamma = 44.02$ against the 41 GeV proton's 43.70, equal to 0.7%, the u-versus- m_p mass difference — where rigidity scaling would give 16.4 GeV/u; the same rule reproduces the published ^3He menu, "41, and 100–183 GeV/nucleon", exactly. One conflict is recorded rather than resolved: Yellow Report Table 10.1 runs 100 GeV protons, $\gamma = 106.6$, in neither stated window. The programme stays anchored on the Yellow Report's three configurations because every beam-parameter table is indexed by them, and `tools/consistency_check.py` keeps the point visible.

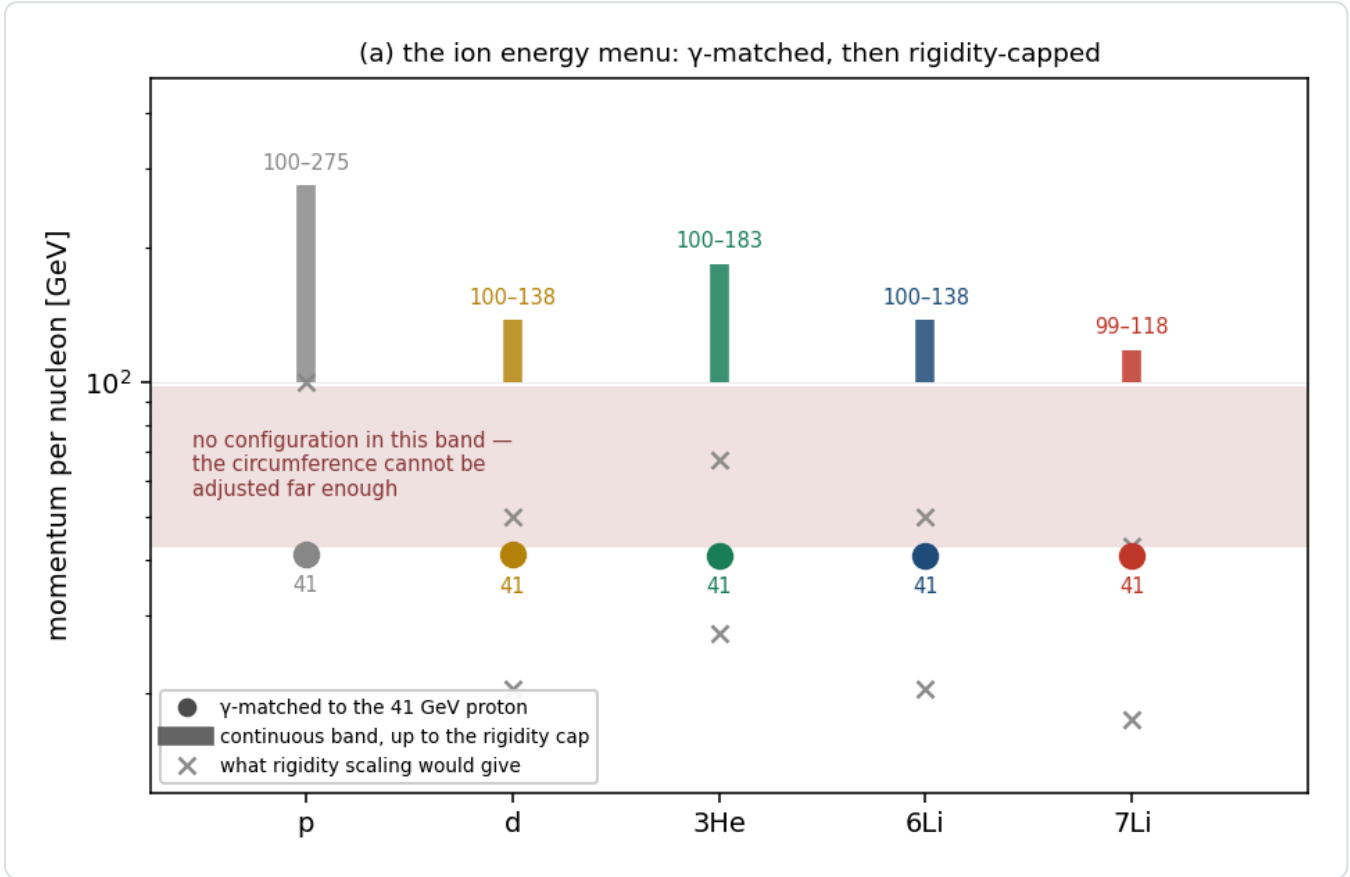


Figure 1 — the ion energy menu. Each species has an isolated γ -matched point at $\approx 41 \text{ GeV/u}$ and a continuous band from $\approx 99 \text{ GeV/u}$ to its own rigidity cap, with nothing between. Crosses mark what rigidity scaling would have given, which this programme used until 2026-08-27. `eic_beam_figures.py`.

SPECIES	A/Z	Γ -MATCHED POINT	CONTINUOUS BAND	RIGIDITY CAP	STATUS
p	1	41 GeV	100–275 GeV	275 GeV	design value
d	2	41.0 GeV/u	100–137.5	137.5	design value
^3He	3/2	40.9	99.8–183.3	183.3	design value (menu published)
^6Li	2	40.8	99.5–137.5	137.5	derived ; top ≈ 138 in EPIOS
^7Li	7/3	40.8	99.5–117.9	117.9	derived ; top ≈ 117 in EPIOS
Au	2.49	41	– (110 published)	110.3	design value

Table 1 — the energy menu. `beams.default_configs` computes these; run on a proton it returns 41 / 100 / 275 exactly, the implementation's self-check. Only the lithium rows are derived, and only their top energies appear in EPIOS [2].

3 · The beams

3.1 · Electron beam

QUANTITY	VALUE	SOURCE	STATUS
energies	5, 10, 18 GeV	YR §10.1	design value
β^* (h/v)	45 / 5.6 cm	EPIOS Table I (EIC column)	design value
RMS emittance (h/v)	20 / 1.3 nm	EPIOS Table I	design value
bunch charge, bunches	28 nC, 1160	EPIOS Table I	design value
divergence h/v, 18 GeV	202 / 187 (HD), 89 / 82 μ rad (HA)	YR Table 10.1	design value
$\Delta p/p$	$5.8\text{--}10.9 \times 10^{-4}$	YR Table 10.1	design value
polarization	70%	YR physics studies	assumption for projections; unused by the tensor observable

Table 2 — the electron beam. The electron divergence enters this programme only through the track angular resolution of Report 2 §3, for which it is the irreducible floor.

3.2 · Hadron beam

Yellow Report Tables 10.1 (e+p) and 10.2 (e+Au) are the authoritative divergence and momentum-spread tables and carry both optics settings; high divergence and high acceptance differ by a β^* choice, and the Yellow Report states that $\sigma_\theta \propto 1/\sqrt{\beta^*}$ while the luminosity goes as $1/\beta^*$, so acceptance is bought with luminosity [1]. The baseline cooling is conventional electron cooling at injection, demonstrated at LEReC; EPIOS is explicit that store cooling is a later-stage addition worth a factor two in average luminosity [2], so the emittance grows under intrabeam scattering through the fill with nothing opposing it. The other design values (EPIOS Table I, EIC column): $\beta^* = 80 / 7.2$ cm, RMS emittance 11 / 1 nm, bunch charge 11.0 nC, 1160 bunches, bunch length 6 cm, peak luminosity $10^{34} \text{ cm}^{-2} \text{ s}^{-1}$. The beam is flat, $\epsilon_x/\epsilon_y \approx 11$, with β^* matched so that the two divergences come out nearly equal — which is why Table 3's h and v agree at 275 and 100 GeV but not at 41.

PROTON CONFIGURATION	HD H/V [MRAD]	HA H/V	$\Delta P/P$ [10^{-4}]	L [$10^{33} \text{ CM}^{-2}\text{S}^{-1}$] HD / HA
275 GeV, 290 bunches	150 / 150	65 / 65	6.8	1.54 / 0.32
275 GeV, 1160 bunches	119 / 119	65 / 65	6.8	10.0 / 3.14
100 GeV	220 / 220	180 / 180	9.7	4.48 / 3.14
100 GeV (vs e 5)	206 / 206	180 / 180	9.7	3.68 / 2.92
41 GeV	220 / 380	220 / 380	10.3	0.44 / 0.44

Table 3 — YR Table 10.1, hadron rows, verbatim (design values). At 41 GeV the two optics are identical: the only configuration with no separate high-acceptance option, and the one the coherent programme would most like one at (§5).

GOLD CONFIGURATION	STRONG HADRON COOLING H/V	$\Delta P/P$	STOCHASTIC COOLING H/V	$\Delta P/P$
110 GeV/u	218 / 379 μ rad	6.2×10^{-4}	77 / 380	10×10^{-4}
41 GeV/u	275 / 377	10×10^{-4}	174 / 302	13×10^{-4}

Table 4 — YR Table 10.2, gold rows. The two cooling scenarios differ by 2.8× horizontally at 110 GeV/u: the cooling choice is worth as much as the optics choice.

4 · Lithium: published, derived, unknown

What is published is thin. EPIOS [2] gives the spin factors and the top energies and states the case; it gives no lithium luminosity, emittance or intensity, and describes the polarized-lithium source as early-stage. Hamwi and Hoffstaetter [3] give the depolarizing-resonance spectrum and the statement that small-G ions cannot use Siberian snakes.

QUANTITY	⁶ LI	⁷ LI	SOURCE / STATUS
spin, A/Z	1, 2	3/2, 7/3	nuclear data
G factor	−0.178 (EPIOS), −0.1818 (Hamwi)	+1.532 / +1.5196	published; the two disagree in the last digits
max Gγ , resonances to cross	27, 81	191, 573	Hamwi Table I, design study
snake scheme	⁶ Li deuteron-like — not amenable to Siberian snakes; partial solenoid and tune jumps; ⁷ Li partial snakes		Hamwi; EPIOS
energy menu	41, and 99–138 GeV/u	41, and 99–118	derived (§2)
σ _θ h/v, high acceptance	220/380, 180/180, 92/92 μrad	220/380, 180/180, 99/99	derived (§4.1)
Δp/p	7–10 × 10 ^{−4}	same	derived; species-insensitive (RF bucket)
tagging optics (§4.2)	β [*] _x × 46 / 164 / 89; envelope 0.36 × 3.8, 0.19 × 1.8, 0.12 × 0.92 mrad; L/L _{HA} = 1/6.8, 1/12.8, 1/9.5	similar	this programme's proposal
emittance, intensity, β [*] , luminosity	– nothing published –		unknown (§7)

Table 5 — the lithium beam.

4.1 · The divergence, derived

σ_θ = √(ε_N/(β_γ β^{*})). At a given machine configuration β^{*} is common to every species, the lattice being set by rigidity, so the only species dependence is through β_γ; and a γ-matched ion has the proton's β_γ exactly (§2). The species factor on the proton divergence is therefore

$$\frac{\sigma_{\theta}^{\text{ion}}}{\sigma_{\theta}^p} = \sqrt{\frac{(\beta\gamma)_p}{(\beta\gamma)_{\text{ion}}}} = 1 \text{ } (\gamma\text{--matched: } 5 \times 41, 10 \times 100), \quad \sqrt{2} \text{ (rigidity capped, } A/Z=2: 18 \times 275)$$

— not a blanket factor, which an earlier version of this programme's correction assumed. Whether equal ε_N is defensible for lithium is tested on gold, the published case: scaling the 275 GeV proton to Au at 110 GeV/u by 1/√(β_γ) alone predicts 236 μrad against the observed 218 (h) and 379 (v), so ε_N(Au)/ε_N(p) = 0.85 horizontally and 2.6 vertically. Intrabeam scattering at fixed beam current goes as Z³/A²: 0.75 for ⁶Li, 1 for a proton and 12.7 for gold — 17× lithium's — and RHIC deuterons showed no measurable IBS growth while gold grew 20–45% per hour. Lithium is a proton-class ion; equal ε_N is the right starting assumption, and it is an assumption. Δp/p is the easy one: protons and gold give 6.2–13 × 10^{−4} across every energy and cooling scheme, set by the RF bucket rather than the species.

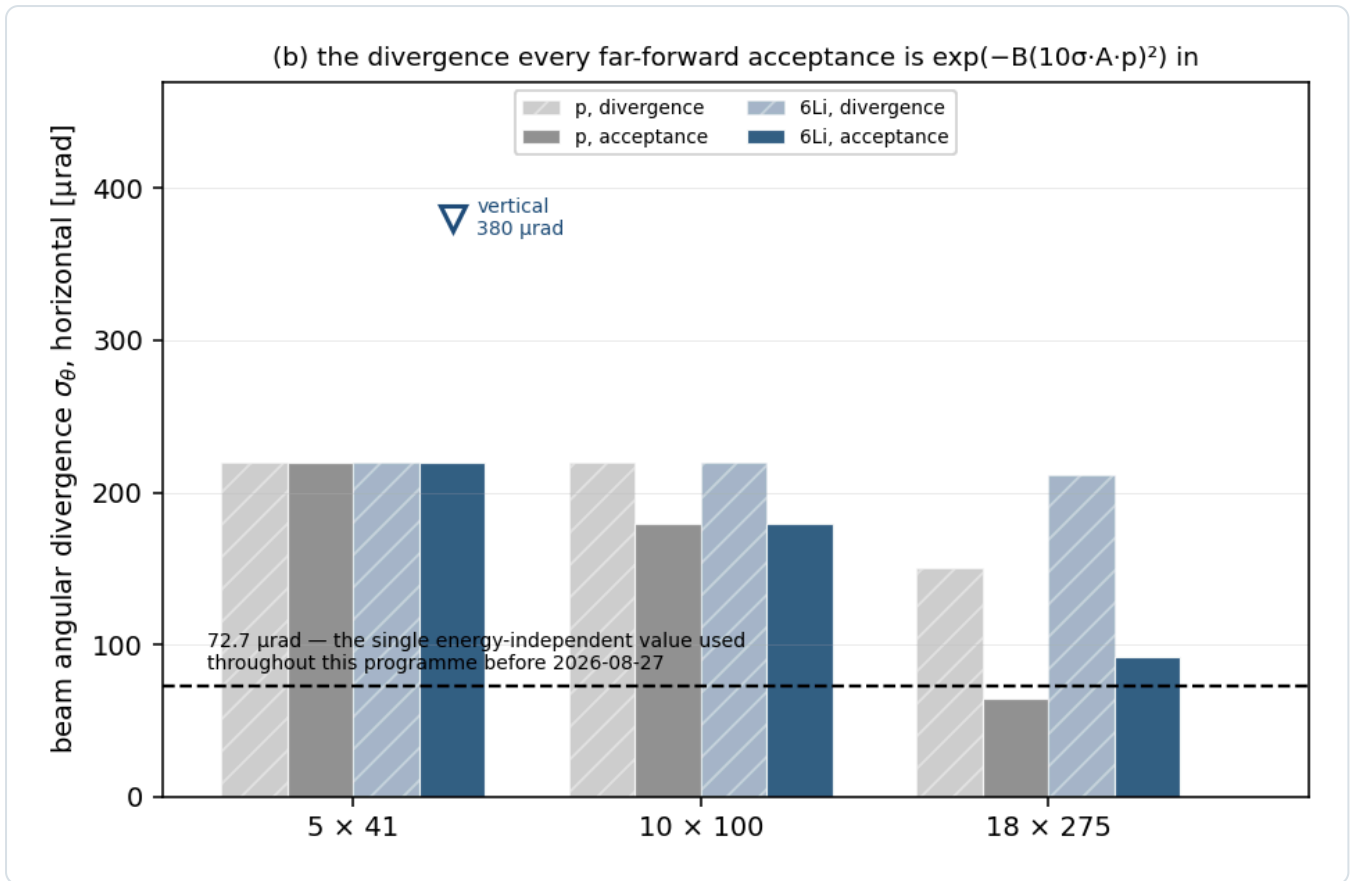


Figure 2 — the divergence. Protons and ^6Li per configuration and optics; the vertical value is marked where it differs from the horizontal. The species $\sqrt{2}$ appears only at 18×275 ; the dashed line is the single energy-independent $72.7 \mu\text{rad}$ used before 2026-08-27, below almost everything. `eic_beam_figures.py`, `farforward.sigma_theta_for`.

4.2 · A lithium tagging optics

The far-forward lithium tags — the intact ^6Li recoil of the coherent channel and the α spectator of the tagged channels — are beam-blind ($A/Z = 2$ exactly; the α at 0.99813 of the beam rigidity) and are separated from the beam only by their angle at the interaction point against the 10σ envelope and the pot aperture. At the optics of Table 3 neither survives (§5). The one lever is β^* : because a recoil escapes through the horizontal gap between the pots or through the vertical one, only the plane it escapes through needs a larger β^* , and Report 1 §6.1 prices the horizontal de-squeeze alone — $\sigma_{\theta,x} \propto 1/\sqrt{r}$ at a luminosity $1/\sqrt{r}$, the vertical plane at high acceptance, the dispersion of the beam's momentum spread at the pots ($D = 0.292$ m against $R_{12} = 29.97$ m at 18×275 , and $0.311 / 19.24$ and $0.287 / 21.25$ at the two lower configurations, all measured on 2026-08-28) added to the horizontal envelope, and the pots following the envelope in both planes. The optimum of tagged fraction $\times$ luminosity is at $r = 46.5$, 164.1 and 89.3 for the three ^6Li configurations, where the envelope is 0.36×3.8 , 0.19×1.8 and 0.12×0.92 mrad, the tagged fraction 0.37, 0.25 and 0.33, and the luminosity $1/6.8$, $1/12.8$ and $1/9.5$ of the high-acceptance value (`farforward.tagging_optics_point`). What that asks of the pots is a number: the horizontal envelope times the measured lever puts the inner edge at 6.99, 4.08 and 3.52 mm from the beam axis at the three configurations, against the 53.6, 31.1 and 26.7 mm the current geometry inserts to (the 2.4 cm base plus the per-energy offsets of §6.1), so the ask is an approach a factor 7.6–7.7 closer than today at all three, and it is the second half of the lever: the β^* de-squeeze without it buys nothing, because the silicon and not the beam would then bind (Report 4 §2). Two assumptions carry it: the electron β^* raised in step to keep the spots matched, and a parallel-to-point far-forward transport for the de-squeezed lattice ($R_{11}\sigma^* \ll R_{12}\sigma_\theta$ at the pots, as in the LHC's high- β^* optics) — true of the present lattice at the Yellow Report optics and the first thing to establish for a de-squeezed one. $\beta_x^* \approx 42$ m at 5×41 ($0.90 \text{ m} \times 46.5$; ≈ 100 m and ≈ 71 m at the other two configurations) is the smallest absolute ask and sits inside the LHC's demonstrated forward-physics range ($\beta^* = 90$ m and 2500 m against a nominal 0.55 m). Unlike the isotropic $\times 14.5$ optimum of plans/10 ($\beta^* = 13$ m), which would have shrunk the beam in the final-focus quadrupoles ($\beta \approx \beta^* + L^2/\beta^* \approx 15$ m against 29 m today for $L \approx 5$ m), a 42 m de-squeeze leaves the horizontal β at the first quadrupole at ≈ 43 m, larger than today, so the IR aperture joins

matching and chromaticity among the questions for C-AD. It is a proposal, not a specification, and the request is stated in §7.

5 • The far-forward lithium tags at each optics

Table 6 evaluates both tags at the Yellow Report optics of each configuration and at the tagging optics of §4.2 with the pots following the envelope. The coherent tag is the analytic acceptance of an $\exp(-B|t|)$ recoil, $B = 50 \text{ GeV}^{-2}$, outside the rectangular cutout, with the pot aperture measured in the ePIC geometry (§6.1) as a second constraint per axis at the Yellow Report optics (cutouts 2.50×6.49 , 1.80×2.12 and 0.92×0.92 mrad, the vertical plane being shut at 5×41); the spectator tags are evaluated against the envelope alone, which binds at 10×99.5 and 18×137.5 and is itself exceeded by the measured 2.50 mrad silicon edge at 5×40.8 , and are the cluster model of `polli_fastsim.spectator` ($\beta = 0.30 \text{ GeV}$, the short-range parameter of the two-parameter cluster momentum density, whose 0.20–0.40 band is the model uncertainty [8]) routed through the far-forward windows of `polli_fastsim.farforward` — Roman Pots at $0.60 \leq R \leq 0.95$, off-momentum detectors at $0.45 \leq R < 0.65$, B0 at 5.5–20 mrad, the near-beam band $|R - 1| < 0.05$ tagged only outside the envelope, and an over-rigid branch at $R > 1.05$ that reaches the Roman-Pot silicon on the inner side of the bend whenever the dispersive displacement clears the per-configuration blind block, 48 / 32 / 16 mm — with the rectangular envelope applied to each fragment's azimuth.

TAG	OPTICS	5 × 40.8	10 × 99.5	18 × 137.5 (⁶ Li) / 18 × 117.9 (⁷ Li)
coherent ⁶ Li recoil, tagged fraction	YR high acceptance + measured aperture	9.4 × 10 ⁻¹⁰	6.2 × 10 ⁻²⁷	7.1 × 10 ⁻¹⁴
	tagging optics, pots follow (L/L _{HA})	0.37 (1/6.8)	0.25 (1/12.8)	0.33 (1/9.5)
⁶ Li α spectator (d struck), tagged	YR high acceptance	0.019	0.017	0.026
	YR high divergence	0.019	0.017	0.025
	tagging optics	0.31	0.22	0.29
⁶ Li d spectator (α struck), tagged	YR high acceptance	0.093	0.081	0.139
	tagging optics	0.54	0.46	0.57
⁷ Li α spectator (t struck), tagged	YR high acceptance	0.969 [†]	0.968	0.975
	tagging optics	0.987	0.991	0.994
⁷ Li t spectator (α struck), tagged	YR high acceptance	0.778	0.919	0.939

Table 6 — the far-forward lithium tags per configuration and optics († the 5 × 40.8 ⁷Li α entry is an angle-only routing; the measured per-band insertion leaves 57%, station 2 only — see below.) (`tagging_acceptance.py` , `nearbeam_aperture_scan.py` ; 2026-08-28). At the Yellow Report optics the ⁶Li α tag is the 1.51, 1.51 and 1.54 points that fall below $R = 0.95$ into the Roman-Pot window, an over-rigid inner branch worth 0.13, 0.16 and 1.00 points at the three configurations, a near-beam tail that clears the envelope in 0.20 points at 5 × 40.8, 0.01 at 10 × 99.5 and 0.08 at 18 × 137.5, and 0.02 points into the B0 at 5 × 40.8 — 1.86, 1.68 and 2.61% in all, which is what the column rounds; the ⁶Li d tag — the deuteron shares the beam's $A/Z = 2$, so it too sits at $R \approx 1$ — is the 5.7–5.9% in the same window plus 1.5, 2.1 and 7.1 points over rigid, a near-beam tail of 1.8, 0.2 and 1.0, and a further 0.3 points into the B0 at 5 × 40.8 alone, with nothing in the off-momentum detectors. The ⁷Li α at $R = 0.856$ lands mid-window regardless of optics at 10 × 99.5 and 18 × 117.9; at 5 × 40.8 the per-energy insertion holds the 32–48 mm silicon band off to $|y| \geq 18 \text{ mm}$, so the same α misses station 1 entirely and reaches silicon only at station 2 and in 57% of events — a module edge, which an angle-only routing cannot see (`tools/fullsim`). The triton ($R = 1.29$) is over-rigid, which since 2026-08-28 is a measured destination rather than a gap: `farforward.route_charged` carries an over-rigid branch (code 6, RP-inner) because a full simulation of the current ePIC geometry puts an $R = 1.286$ triton on the Roman-Pot silicon in 60 of 60 events at every configuration, at $dx = +66 \text{ mm}$ on the inner side of the bend, followed by a deposit in the zero-degree calorimeter in 80–98% of the same events (`tools/fullsim` , plans/09 B1 [8]). The ⁷Li α + t double tag is therefore available at IP6, as an RP-inner track plus calorimeter energy, and the triton row above is that branch and not the low-momentum tail: it is dispersive rather than angular, so the tagging optics moves it by under 0.3 points, to 0.780 / 0.922 / 0.941. What the double tag does not yet have is the α + t overlap with its m-dependence, one of the theory inputs collected in `docs/note_7li_theory_questions.md` . The third column is the top configuration of each isotope, 137.5 GeV/u for ⁶Li and the 117.9 GeV/u rigidity cap for ⁷Li. Before 2026-08-28 the fast simulation applied a single proton-derived 73 μrad at every configuration, and applied it azimuth-blind as a circle: that pair of

choices gives 2.8% for the ${}^6\text{Li}$ α at 18×137.5 and 16.8% at 5×40.8 , against 2.6% and 1.9% here; before the over-rigid branch entered every column the same comparison read 1.9% and 20% against 1.6% and 1.7%, and 1.85% is what was published. Two readings of the ${}^6\text{Li}$ α number coexist in this repository and neither stands in for the other. The rows above are `polli_fastsim.spectator`, one partial wave per channel — $L = 0$ for α -d — while the tagged generator behind Report 4 §2.1 (`polligen.tagged`) carries the full $S + D$ expansion of the same channel: the S -wave radial is identical in the two, $\langle k \rangle = 0.1071$ GeV/c, and the D wave is hard at $\langle k \rangle = 0.2778$, so at $P_D = 0.0867$ the channel mean moves to 0.1219 and the off-rigidity slice from 1.5% to 2.5%. The D wave is not an optional tail but the tensor observable itself — set $P_D = 0$ and the α -d density is m -independent and A_{zz}^{tag} vanishes identically — so the tagged asymmetries are quoted on the $S + D$ spectrum and this table on the S -wave one: the Roman-Pot slice of the rows above is 1.5% at every configuration, against 2.5–2.9% there. The two ${}^7\text{Li}$ densities are the same pure P wave, and the 0.7 points by which this table's 0.9690 at 5×40.8 exceeds the tagged script's 0.9620 is a difference of *acceptance definition*: the column above is 1 — lost, i.e. any far-forward system, while `tagged_polarimetry_7li.py` masks on the Roman Pots alone, and at 5×40.8 the B0 carries 1.1% of the ${}^7\text{Li}$ α and the over-rigid inner branch a further 0.2%. Appendix A carries the like-for-like comparison.

Two conclusions. The tagging inversion between isotopes at IP6 is optics-independent: the ${}^7\text{Li}$ α works at 97% at every optics — in angle; at 5×40.8 the measured per-band insertion leaves 57%, station 2 only, and the 0.969 of Table 6 is an angle-only routing there (caption) — while the ${}^6\text{Li}$ α does not work at any published one. And the tagging optics converts the ${}^6\text{Li}$ tags from beam-blind to usable — 25–37% of coherent recoils, 22–31% of α spectators, 46–57% of deuteron spectators — at a luminosity cost of 6.8–12.8, which is the trade Report 1 §6.1 prices and Report 4 examines from the detector side. One caution on reading the two ${}^6\text{Li}$ spectator rows: they are not independent channels but the two fragments of the same breakup, and sampled jointly (Report 4 §4.2) both are recorded in 22–29% of breakups at the tagging optics — so four fifths of the α row there lies inside the deuteron row, while about half the deuteron row is deuteron-only, the deuteron being the systematically wider fragment; at the Yellow Report optics, where the joint fraction is below 0.03%, the two rows are effectively disjoint.

The luminosity cost is not shared evenly between the isotopes, and for ${}^7\text{Li}$ it is not paid for at all. The ${}^7\text{Li}$ α row moves from 0.968–0.975 to 0.987–0.994 under the tagging optics — $\times 1.02$ — for a luminosity of 1/7.9, 1/14.8 and 1/10.1, so the de-squeeze is a factor 7.9–14.8 *net loss* in tagged ${}^7\text{Li}$ events and multiplies every ${}^7\text{Li}$ error bar by 2.78, 3.81 and 3.15 at equal running time (Report 4 §2.1). The α tag that lands mid-window is the one the machine cannot improve. So the inversion in the first conclusion runs in both directions: ${}^6\text{Li}$ needs the de-squeeze and ${}^7\text{Li}$ must not have it, which makes the two isotopes separate runs with separate optics rather than one fill plan — a scheduling consequence to put to C-AD alongside the lattice questions of §7.

6 • The ePIC detector

6.1 • Far forward

SUBSYSTEM	POSITION	ACCEPTANCE	TECHNOLOGY	STATUS
B0	$z \approx 5.4\text{--}6.4$ m	$5.5 < \theta < 20$ mrad, any charged; γ	AC-LGAD + EMCal	design value
off-momentum detectors	$z = 25.5,$ 27.0 m	$\theta < 5$ mrad, $0.45 \leq R < 0.65$	AC-LGAD	design value (windows [6]); positions from the geometry file
Roman Pots	$z = 32.55,$ 34.25 m	$\theta < 5$ mrad, $0.60 \leq R \leq 0.95$; inner edge optics-dependent	AC-LGAD, in vacuum	geometry file (current main)
ZDC	$z \approx 37.5$ m	$\theta < 4$ mrad, neutrals	W/Si + crystal	design value

Table 7 — the far-forward systems. The pot and off-momentum positions are read from `compact/far_forward/roman_pots_eRD24_design.xml` and `offM_tracker.xml` on the current main branch [4] and are not the 26 / 28 and 22.5 / 24.5 m the Yellow Report, the ePIC wiki and several papers still quote; only the angular windows enter the routing.

The pot aperture, measured. An intact ${}^6\text{Li}$ shot through the ePIC far-forward geometry (`tools/fullsim`, `epic-main` at git 9aaa2969, 216 map points in $p_T \times$ azimuth plus four 0.05 mrad angular ladders of 468 points each per configuration, one event per point, no beam envelope) clears the pots at $|\theta_x| \gtrsim 2.50 / 1.51 / 0.53$ mrad in the $5 \times 41 / 10 \times 100 / 18 \times 275$

optics against $|\theta_y| \gtrsim$ (no vertical acceptance) / 2.12 / 0.92 mrad: the aperture opens horizontally, because the beamline images an IP angle onto the pot plane with a horizontal lever $R_{12} = 19.24 / 21.25 / 29.97$ m against a vertical R_{34} of 4.56 / 3.35 / 2.93 m — a factor 4.2, 6.3 and 10.2 — the 5×41 entry measured on 2026-08-29 through a scratch geometry with the four Roman-pot section offsets zeroed, the real insertion there shutting the vertical plane. The 16×16 mm modules and per-energy 10σ offsets of the current geometry are what these numbers are measured in; against the September-2024 reading the aperture halved at 18×275 and widened at 5×41 , which reverses the ordering of §5: the silicon binds at 5×41 and the machine at 18×275 , where the earlier geometry said the opposite. The 5×41 row carries one further condition, and it is the largest uncertainty in this paragraph: it stands on the ePIC baseline compact files, whose 5×41 lattice is the separately re-matched proton one, taken here — as an educated guess stated in §7 — to be the lattice a ${}^6\text{Li}$ fill at 81.6 GV would run in at twice the field, while the $Z/A = 0.5$ light-ion lattice carried as the alternative gives $R_{12} = 29.81$ m and a 1.61 mrad edge — 48 mm / 29.81 m, the threshold the code applies, the first hit landing at 1.60 mrad on +x and 1.70 on -x — a factor 1.55 in the lever and 0.64 in the aperture (`farforward.POT_LEVERS_LIGHT_ION_LATTICE` ; §7). Status: **measured, current geometry (2026-08-28)**.

Sensor and readout. AC-LGAD at 500×500 μm pitch with a 100×100 μm metal electrode over a 30 μm active thickness, bump-bonded to EICROC; timing ≈ 35 ps per plane is a requirement, HPK AC-LGADs measure 20–35 ps in test beam; spatial resolution ≈ 140 μm from the pitch alone, ≈ 20 μm shown with charge sharing, ePIC's stated pot baseline binary 500 μm without it; EICROC provides a 10-bit 25 ps time-of-arrival TDC and an 8-bit 40 MHz SAR ADC for charge [5]. The insertion is quantised by 16×16 mm modules with a 2.4 cm base plus a per-energy offset (0.27 / 0.71 / 2.96 cm at 18×275 / 10×100 / 5×41), with the geometry file's own comment that these "are the ten-sigma cuts for the Roman pots, translated to the physical layout we currently have" [4]. Rates: a few Hz per channel in normal running, 30–50 kHz at $\approx 10\sigma$ from beam halo on STAR experience [7]; in-vacuum cooling ≈ 100 W per sensor layer; no published radiation fluence or dose.

6.2 • Central detector

QUANTITY	VALUE USED	SOURCE	STATUS
EMCal σ_E/E , backward ($\eta < -2$)	2%/√E $\oplus$ 1%	YR Fig. 8.3; the PbWO_4 crystal expectation	requirement, correctly sourced
... backward transition ($-2 < \eta < -1$)	7%/√E $\oplus$ 1%	YR Fig. 8.3	stochastic exact; constant 2× optimistic against YR Table 3.1
... barrel ($ \eta < 1$)	10%/√E $\oplus$ 1%	YR Table 8.20 band (10–12)%/√E $\oplus$ (1–3)%	optimistic corner; constant 2–3× optimistic against the ePIC BIC requirement
... forward ($\eta > 1$)	7%/√E $\oplus$ 1%	—	mis-sourced: 7%/√E belongs to $-2 < \eta < -1$; latent, no sweet-spot electron is forward
... noise term	absent	YR Table 3.1 has 1–2%/E	missing; 2% at 1 GeV
tracking σ_p/p	$\sqrt{((a/p)^2 + b^2)}$, η -piecewise	YR Fig. 8.3 / Table 11.2; ATHENA Table 4 quotes the same as "Requirements"	requirement — barrel and forward exact, endcap constants padded; ePIC full simulation reaches 0.45–0.6% at 1 GeV/c, so realistic to $\approx \pm 20\%$
tracking σ_θ	5 / 3 / 2 / 1 / 2 / 3 / 5 mrad by η	—	placeholder — no angular requirement in either YR table; also the model's only stand-in for the 81–211 μrad beam-divergence floor on θ' , so the honest bracket is a factor ≤ 3 (Report 2 §3)
electron identification	0.70–0.95 by η	constructed between ATHENA JINST 17 P10019 Table 5 and ECCE NIM A 1055 168464 §3.5.2	anchored at one point; the sources differ by ≈ 25 points in the barrel; cancels in the spin-state ratio
hadron-side coverage,	tracks $ \eta \leq 3.5$, $p_T > 0.2$ GeV, 95%;	—	stand-ins of the right magnitude (Report 2 Table 2); the ePIC nominal forward reach is

thresholds, efficiencies, noise	calorimeters $ \eta \leq 3.7$; 50 MeV noise	4.0
hadron-side resolutions	YR tables per region, 2% EMCal constant	YR requirement tables requirement

Table 8 — the central-detector model, fact-checked against the Yellow Report. Every sweet spot of the programme puts the scattered electron backward ($\eta = -2.9$ to -1.6), where the calorimeter table is right to within 20% and where the tracker is the better measurement anyway; the mis-sourced forward row is latent. The angular resolution is the entry to watch: the 0.1 mrad quoted in some ePIC talks is a smearing input, not a measured resolution, and would delete the beam-divergence floor the placeholder stands in for.

7 • What the programme assumes that the specification does not give it

#	WHAT IS MISSING	WHAT WE DO INSTEAD	OWNER
1	lithium beam divergence and $\Delta p/p$, per configuration and cooling scenario	derive from the proton tables with the γ -matched species step (§4.1)	C-AD / ePIC FF WG
2	a lithium tagging optics: can IR6 be matched with the horizontal hadron β^* raised 46–164 $\times$ — the electron β^* co-de-squeezed so ξ_p stays ≤ 0.015 — and what are the IP-to-pot phase advances (parallel-to-point transport)?	assume the optimum of §4.2 exists; without it no ^6Li far-forward tag exists at IP6 (§5)	C-AD
3	lithium emittance, intensity, β^* and luminosity — nothing published	assume proton-class ϵ_N (gold calibration, Z^3/A^2 IBS scaling) and the Yellow Report per-nucleon luminosity	C-AD / ANL source group
4	what physically sets the 10σ retraction — collimation hierarchy, halo lifetime, or convention — and pots that follow the envelope of a de-squeezed optics	assume 10σ and pots that move (§4.2, Report 4 §3)	C-AD / ePIC FF WG
5	EICROC input charge dynamic range in fC, and LGAD gain suppression at ≈ 9 MIP	assume the charge is usable; the far-forward Z-identification depends on it entirely (Report 4 §4)	OMEGA- IJCLab / BNL AC-LGAD group
6	per-optics transfer matrix at the pot planes — R_{11} , R_{12} , R_{21} , R_{22} , D , D' , and which 5×41 lattice a ^6Li fill runs in	use the measured (R_{12} , R_{34} , D) = 19.24 / 4.56 / 0.311, 21.25 / 3.35 / 0.287 and 29.97 / 2.93 / 0.292 m; the 5×41 R_{34} was obtained on 2026-08-29 through a scratch geometry whose four Roman-pot section offsets are zeroed, the real insertion there shutting the vertical plane, and R_{11} , R_{21} , R_{22} and D' are still unmeasured. The 5×41 triple is the ePIC baseline (proton) lattice, which we adopt for the ^6Li fill as an educated guess — the same magnets and orbit at twice the field for the $Z/A = \frac{1}{2}$ beam — with the $Z/A = 0.5$ file at the same ring setting (29.81 m, $\times 1.55$ in the lever and $\times 0.64$ in the aperture) carried as the labelled alternative. What is asked here is confirmation of that choice, not a decision the published numbers wait on	ePIC FF WG
7	ePIC tracking angular resolution and electron identification — no published curves	η -piecewise stand-ins, flagged in the code and in Table 8	ePIC
8	the calorimeter noise and threshold floor on the hadronic $E - p_z$ sum at $\Sigma_h = 0.2$ –0.5 GeV	50 MeV Gaussian stand-in; it sets the $y \approx 0.01$ bins (Report 2 §4.2)	ePIC inclusive WG
9	radiation environment at the Roman Pots — no published	not modelled	ePIC FF WG

	fluence or dose		
10	polarization-preservation scheme for ${}^6\text{Li}$, which cannot use Siberian snakes	assume it works at the EPIOS source targets	C-AD / EPIOS
11	no EIC document specifies a tensor-polarization scale uncertainty, for any species; the 1% requirement is vector-only	ours to specify: $\delta P_{zz}/P_{zz} \leq 5\%$ (Report 2 §6), a 1:1 normalisation on Δ	ANL source + BNL polarimetry
12	relative luminosity between fills of different P_{zz} is unspecified; the $R < 10^{-4}$ requirement is bunch-by-bunch within a fill	ours to specify: $\leq 10^{-3}$ on the ratio, or bunch-by-bunch alternation of the tensor states, which also makes the acceptance cancellation exact	EIC project / ePIC luminosity / source
13	EICrecon has no light-ion far-forward configuration: the transfer matrix is selected by matching the beam proton PDG code	this repository does its own transport; the deliverable is a beamline configuration and a transfer matrix EICrecon can adopt	this programme
14	Roman-Pot commissioning is deferred past year 1 and conditional on beam conditions	projections assume the pots exist from the start	ePIC
15	a 9 GeV electron beam is the early-science baseline; no ESR lattice, emittance, β^* or polarization table exists for it	only 5, 10 and 18 GeV are modelled	C-AD
16	the run plan: how a year of running divides between isotopes, ion fills and far-forward optics — no EIC document allocates it, and the Yellow Report luminosity is quoted per configuration, not per measurement	ours to propose (plans/07 WP2): every projection is quoted at the full year in its own configuration, and a share f of that year multiplies every statistical error by $1/\sqrt{f}$, leaves a luminosity quoted as a reach where it is, and multiplies the years to that reach by $1/f$	this programme / EIC project

Table 9 — the assumptions, ranked by leverage on a published number. Items 1–10 and 14–15 are machine and detector inputs, to be asked for precisely; 11, 12 and 16 are properties of a measurement nobody has designed yet and are this programme's to specify; 13 is the point of the development rather than a gap in the specification.

8 • Summary

The electron and proton beams are specified in detail and this programme uses them as specified; the gold rows calibrate the one derivation that matters, the lithium divergence; the ion energies follow from the revolution-period constraint and are checked against the published gold and ${}^3\text{He}$ menus. For lithium itself essentially nothing exists beyond a top energy and a G factor, so §4 is a derivation and not a citation, and its consequence is the sharpest statement this report makes: at every published optics the intact ${}^6\text{Li}$ recoil and the ${}^6\text{Li}$ α spectator are inside the beam envelope, and the far-forward half of the polarized-lithium programme exists only with a tagging optics — a one-plane β^* de-squeeze of 46–164 with pots that follow it, at $1/6.8$ – $1/12.8$ of the luminosity — which is the request to put to C-AD. The detector is specified where it matters most for the tags, in a geometry file that can be read, and least where the central reconstruction lives, which is why Table 8 marks the track angular resolution a placeholder and the electron-identification curve a construction between two non-ePIC sources, rather than giving numbers that would look like a specification.

References

- [1] R. Abdul Khalek et al., *Science requirements and detector concepts for the Electron-Ion Collider: EIC Yellow Report*, Nucl. Phys. A 1026 (2022) 122447, arXiv:2103.05419 (refs/2103.05419_part1.pdf ... refs/2103.05419_part4.pdf): Tables 10.1 and 10.2

- (beam parameters per configuration and optics), Table 11.48 (Roman-Pot p_T smearing), §11.6 (far forward), §14.5.3 (superconducting nanowires), and the detector requirement tables.
- [2] G. Atoian et al. (EPIOS), *Realizing the scientific program with polarized ion beams at EIC*, Phys. Rev. C 113 (2026) 060501, arXiv:2510.10794 (refs/2510.10794.pdf): Table I (HERA versus EIC parameters); the species table with G and mc^2/G ; the synchronisation mechanism (pp. 12–13); the polarized-lithium case.
- [3] E. Hamwi, G. H. Hoffstaetter, *Polarization transmission in the Electron-Ion Collider's Hadron Storage Ring*, Phys. Rev. Accel. Beams 29 (2026) 073501, arXiv:2509.18558 (refs/2509.18558.pdf): Table I, the species-dependent resonance spectrum; small-G ions cannot use Siberian snakes.
- [4] eic/epic, branch main: compact/far_forward/roman_pots_eRD24_design.xml , offM_tracker.xml and compact/fields/beamline_*.xml , read directly 2026-08-26/27: station positions, module size, per-energy insertions, the RF-shield comment.
- [5] A. Jentsch (ePIC), *Far-Forward Detectors and Physics with ePIC @ the EIC*, DIS 2023 (refs/ePIC_far_forward_talk_DIS_2023_v2.pdf), and the ePIC Collaboration Meeting, JLab, July 2025: acceptances, the AC-LGAD specification, the in-vacuum cooling load, the x-moving layer.
- [6] A. Jentsch, Z. Tu, C. Weiss, Phys. Rev. C 104 (2021) 065205, arXiv:2108.08314 (refs/2108.08314.pdf): Table I, the far-forward acceptance windows the router uses.
- [7] ePIC Collaboration, *Preliminary Design Report*, September 2024: Roman-Pot rates, a few Hz per channel in normal running and 30–50 kHz at $\approx 10\sigma$ from beam halo.
- [8] This repository: Report 1 §6.1 and evgen/scripts/tagging_optics.py (the tagging optics priced), Report 2 (the reconstruction chain), Report 4 (the near-beam study), plans/10_beam_divergence_light_ions.md (the derivation record), fastsim/scripts/tagging_acceptance.py (Table 6), plans/05_doubly_polarized_generator.md §5.2 (the cluster momentum density and its β band), plans/03_phase2_full_simulation.md §2.2 and tools/fullsim/README.md (the far-forward routing and the particle-gun scan).

Appendix A · Revision history

DATE	REVISION
2026-08-29	Verification pass over the round. §5's first conclusion and Table 6's ${}^7\text{Li}$ α entry carry the 5×40.8 caveat the caption already stated: the 97% is an angle-only routing, and the measured per-band insertion leaves 57%, station 2 only. Table 6's caption decomposes the ${}^6\text{Li}$ α tag per configuration — 1.51 / 1.51 / 1.54 points in the Roman-Pot window, 0.13 / 0.16 / 1.00 over rigid, 0.20 / 0.01 / 0.08 in the near-beam tail and 0.02 into the B0 at 5×40.8 — which sums to the column rather than to a single 1.5%. §6.1's lever ratio is quoted per configuration, 4.2, 6.3 and 10.2, in place of a factor 9; the $Z/A = 0.5$ lattice edge as the applied 1.61 mrad; the ${}^7\text{Li}$ error-bar penalty as 2.78, 3.81 and 3.15, on the per-configuration tagging levers of that day. The 2026-08-28 row's "1.6% and 1.7%" is labelled as the pre-over-rigid reading. Later the same day the per-configuration transport became the baseline for the tagging optics as well as for the coherent channel, and Table 6's three tagging rows were re-run on it: the ${}^6\text{Li}$ α tag is 0.31 / 0.22 / 0.29 against 0.35 / 0.28 / 0.29, the ${}^6\text{Li}$ d tag 0.54 / 0.46 / 0.57 against 0.57 / 0.51 / 0.57 and the ${}^7\text{Li}$ α tag 0.987 / 0.991 / 0.994 against 0.988 / 0.992 / 0.994, with the abstract, §5 and the derived ranges (22–31% and 46–57%) following. §4.2 now states what the optics asks of the pots in millimetres — an inner edge at 6.99, 4.08 and 3.52 mm against the 53.6, 31.1 and 26.7 mm inserted today — rather than leaving the pot half of the lever qualitative. The 5×41 vertical lever $R_{34} = 4.56$ m, measured through a zeroed-insertion scratch geometry, enters §6.1 and Table 9; and the choice of 5×41 lattice is stated as a choice — the ePIC baseline file at twice the field, an educated guess, with the $Z/A = 0.5$ file as the labelled alternative — so Table 9 row 6 asks the far-forward working group to confirm it rather than to settle a fork the numbers wait on.
2026-08-28	Table 9 gained item 16, the run plan: no EIC document allocates a year of running between isotopes, ion fills and far-forward optics, and the Yellow Report luminosity is quoted per configuration rather than per measurement, so the allocation is this programme's to propose (plans/07 WP2). Every projection in this series is quoted at the full year in its own configuration; under a share f of that year every statistical error grows as $1/\sqrt{f}$, a luminosity quoted as a reach does not move, and the years needed to accumulate it grow as $1/f$.
2026-08-28	Rewritten as a paper. The far-forward optics are now per configuration throughout the code (farforward.yr_optics , tagging_optics , a rectangular envelope applied to each fragment's azimuth in the spectator routing — at every aspect ratio since a 2026-08-28 review, an isotropic envelope having until then fallen back on its inscribed circle), and §5 with Table 6 is new: the lithium tags at the Yellow Report optics and at the tagging optics of Report 1 §6.1, replacing the single proton-derived 73 μrad the fast simulation had applied at every configuration, which gave the ${}^6\text{Li}$ α tag as 1.85% at 18×137.5 and 20% at 5×40.8 against the 1.6% and 1.7% this section read before the over-rigid branch entered every column later the same day; the current Table 6 reads 2.6% and 1.9%. §4.2 states the tagging optics as a proposal; §7 adds the two questions it raises for C-AD and the calorimeter noise floor. $\beta = 0.30$ GeV is defined where it is first used, and Table 6's triton row is now the measured over-rigid branch. §5 gained the ${}^7\text{Li}$ luminosity-cost paragraph — the tagging optics is a factor 8–15 net loss for ${}^7\text{Li}$, so the two isotopes are separate runs — and Table 6's caption the two-sampler note: S wave here, S + D in Report 4 §2.1, 1.5% against 2.5–2.9% (plans/09 B2, B3). The ${}^7\text{Li}$ residual between the two samplers is a difference of acceptance definition and not of density, and like for like they agree to 0.1 point: the pure model's Roman-Pot tag is 0.9568 / 0.9668 / 0.9721, and inside the $k \leq$

1.2 GeV/c on which the tagged model's momentum grid ends 0.9626 / 0.9683 / 0.9736, against the tagged sampler's 0.9620 / 0.9678 / 0.9728; at 10×99.5 and 18×117.9 the B0 fraction is zero and the two definitions coincide. With $P_D = 0$ the two ${}^6\text{Li}$ samplers agree quantile by quantile. Late the same day the pot aperture was re-measured in the current ePIC geometry (`epic-main` at git 9aaa2969), and §6.1, §7 and every row of Table 6 follow it: the horizontal edge is 2.50 / 1.51 / 0.53 mrad against 2.00 / 1.35 / 1.03, the levers (R_{12} , R_{34} , D) are measured per configuration for the first time, the coherent tag at the Yellow Report optics moves to 9.4×10^{-10} / 6.2×10^{-27} / 7.1×10^{-14} , and the over-rigid triton — 60 of 60 events on Roman-Pot silicon at $dx = +66$ mm with a ZDC deposit behind it — lifts the ${}^7\text{Li}$ t tag from 0.033 / 0.004 / 0.005 to 0.778 / 0.919 / 0.939, retires "no coverage" and opens the ${}^7\text{Li}$ $\alpha + t$ double tag. The ${}^6\text{Li}$ α tag rises to 0.019 / 0.017 / 0.026 and the d tag to 0.093 / 0.081 / 0.139 through the same branch, applied against the per-configuration 48 / 32 / 16 mm blind block (plans/09 B1, `tools/fullsim/README.md`).

2026-08-27 First version, after the γ -matching correction (plans/10): the ion energy menu, the beam tables, the lithium divergence derived, the far-forward geometry read from the current main branch, the central-detector table fact-checked against the Yellow Report (the tracking resolution is the YR requirement and not a placeholder; the calorimeter forward row is mis-sourced; the noise term is missing), and the list of unknowns with owners. Later the same day: the synchronisation mechanism verbatim from EPIOS and the conflict with Table 10.1's 100 GeV row; store cooling not in the baseline.

Status vocabulary: **design value** — appears in a design document as a parameter; **requirement** — a specification the subsystem must meet; **measured** — test beam, operations, or a simulation of the geometry; **derived** — computed here from published inputs; **unknown** — not published anywhere we could reach. Beam divergences and $\Delta p/p$ are Yellow Report Tables 10.1 and 10.2 unless stated; ion energies are computed by `fastsim/polli_fastsim/beams.py` and pinned by `fastsim/tests/test_beams.py` ; the far-forward optics by `polli_fastsim/farforward.py` and `fastsim/tests/test_optics_20260828.py` . Built by `reports/build_report.py` .